Chapter 2

Table of contents

Spatial Point Pattern

Background, Definition, Application Areas and SPP Data Types

Background, Definition

- A 'spatial point process' is a stochastic process in which we observe spatial locations of some things or events of interest within a bounded region A.
- The locations of the events generated by a point process in the area of study A will be called a point pattern.
- The spatial arrangement of points is the main focus of investigation.

Ref: [3] page 3. Figure 1.1

Point pattern analysis is concerned with describing patterns of points over space and making inference about the process that could have generated an observed pattern. The main focus here lies on the information carried in the locations of the points, and typically these locations are not controlled by sampling but a result of a process we are interested in, such as animal sightings, accidents, disease cases, or tree locations.

In terms of random processes, in point processes locations are random variables, where, in geostatistical processes the measured variable is a random field with locations fixed.

Point pattern datasets with spatially varying density of points. Left: enterochromaffinlike cells in histological section of gastric mucosa (T. Bendtsen; see [484, pp. 2, 169]), interior of stomach towards top of picture. Right: sky positions of 4215 galaxies in the Shapley Supercluster (M. Drinkwater); survey region about 25 degrees across.

Application Areas

Examples include:

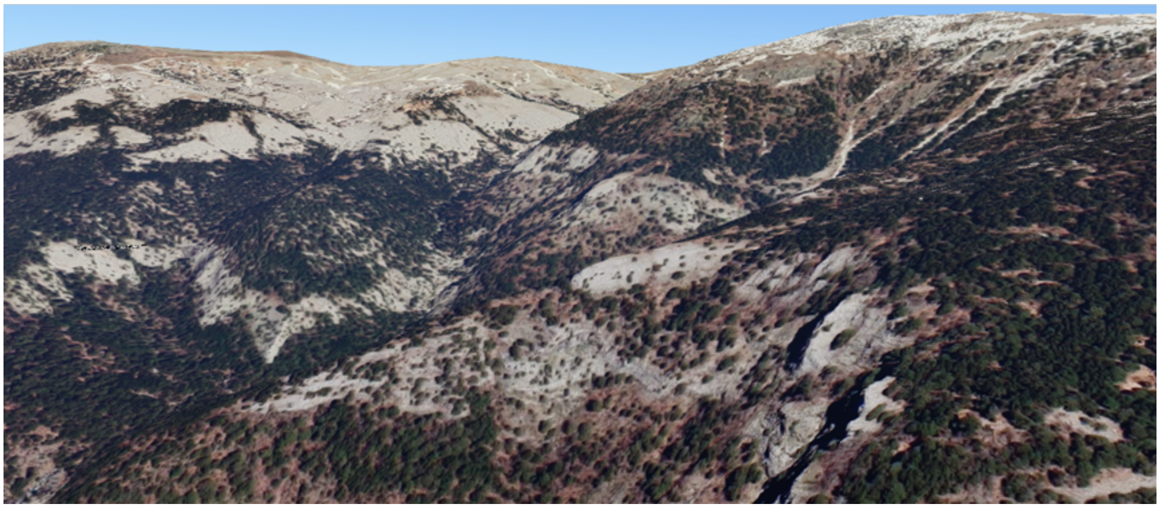
- the locations of trees in a forest, the spatial distribution of the trees, competition or coordination between different species of vegetation
- gold deposits mapped in a geological survey,
- stars in a star cluster,
- road accidents,
- earthquake epicentres,
- mobile phone calls,
- facility locations,
- animal sightings,
- cases of a rare disease, are they clustered, reasons why,
- place where crime occurs – hotspots (we need good data for this: check out:

Examples in the literature span a wide range of application areas, with many originating in forestry and ecology. Within the social sciences, however, applications are most commonly found in criminology, where researchers examine the spatial distribution of crimes, accidents, and other incidents. In economic geography, point pattern analysis has been used to investigate the distribution of facilities and settlements to determine whether there are meaningful spatial relationships between them.

As with many forms of spatial analysis, the starting point is the assumption of spatial randomness. Our goal is typically to test this assumption and, where possible, reject it in favor of evidence of a systematic spatial pattern. Once such patterns have been identified, we can begin to investigate and explain the processes that may have produced them. The same principle applies to point pattern analysis.

Let us now consider some examples.

Example 1: Trees



In this picture (from google maps 3D), you see an example for a point process from Kaz Dağı, the ancient Mount Ida, in the famous epic poem (The Iliad) of Homer (eighth-century BC), the mountain was dedicated to gods and goddess, and all kinds of divine happenings took place on this mountain according to the poem. It is a mountainous area and one of the important biodiversity spots, located in the north of Edremit in northwestern Anatolia.

This picture from Google depicts the high mountain vegetation in the foothills of Sarikiz hill (1726m) which is the spot where the beauty contest took place (Judgement of Paris). As you go to lower altitudes, you will see Kaz Dagi Fir (1000-1400m), Black pine forests (700-750m), Turkish red pine (250-300m), and lots of researchers examine the reasons why the decline in the vegetation. Climate Change Impacts on High Altitude Ecosystems – Chapter 13 (page 343-358) is an example.

In mapping a snapshot of the forest, we hope to understand ecological processes, such as competition for resources (soil nutrients, light, water, growing space), and spatial variation in the landscape, such as variation in soil type or soil fertility.

Example 2: Cholera

Example 3: Stars – Ara Constellation

Ref: https://www.constellation-guide.com/constellation-list/ara-constellation/ Ara was one of the 48 Greek constellations listed by the astronomer Claudius Ptolemy in the 2nd century. Ara contains two stars brighter than magnitude 3.00 and three stars located within 10 parsecs (32.6 light years) of Earth.

Example 4: Clinics

Download the data from the following:

https://web1.capetown.gov.za/web1/OpenDataPortal/DatasetDetail?DatasetName=Clinics

```r
library(sf)
```

```
Warning: package 'sf' was built under R version 4.4.3
```

```
Linking to GEOS 3.13.0, GDAL 3.10.1, PROJ 9.5.1; sf_use_s2() is TRUE
```

```r
RSA_roads = "C:/Users/01438475/OneDrive - University of Cape Town/ASDA/DataSets/zafrds8ff5a/Z
  st_read() |> st_transform(4326)
```

```
Reading layer `ZAF_roads' from data source
  `C:\Users\01438475\OneDrive - University of Cape Town\ASDA\DataSets\zafrds8ff5a\ZAF_roads.s
  using driver `ESRI Shapefile'
Simple feature collection with 5559 features and 5 fields
Geometry type: MULTILINESTRING
Dimension:     XY
Bounding box:  xmin: 16.48784 ymin: -34.82747 xmax: 32.85267 ymax: -22.17693
Geodetic CRS:  WGS 84
```

```r
RSA_biome = "C:/Users/01438475/OneDrive - University of Cape Town/ASDA/DataSets/rsabiome4xkhi
  st_read() |> st_transform(4326)
```

```
Reading layer `RSA_biome' from data source
  `C:\Users\01438475\OneDrive - University of Cape Town\ASDA\DataSets\rsabiome4xkhi\RSA_biome
  using driver `ESRI Shapefile'
Simple feature collection with 3127 features and 1 field
Geometry type: POLYGON
Dimension:     XY
Bounding box:  xmin: 16.45296 ymin: -34.8336 xmax: 32.8928 ymax: -22.12583
Geodetic CRS:  Hartebeesthoek94
```

```r
clinics_sf = "C:/Users/01438475/OneDrive - University of Cape Town/ASDA/DataSets/Clinics/SL_C
  st_read() |> st_transform(4326)
```

```
Reading layer `SL_CLNC' from data source
  `C:\Users\01438475\OneDrive - University of Cape Town\ASDA\DataSets\Clinics\SL_CLNC.shp'
  using driver `ESRI Shapefile'
Simple feature collection with 149 features and 5 fields
Geometry type: POINT
Dimension:     XY
Bounding box:  xmin: 18.34268 ymin: -34.19491 xmax: 18.90847 ymax: -33.51262
Geodetic CRS:  WGS 84
```

```r
ct.wards_sf <- "C:/Users/01438475/OneDrive - University of Cape Town/ASDA/DataSets/sa/CPT/ele
  st_read(quiet = TRUE) |>
  st_set_crs(4326)

st_crs(ct.wards_sf)
```

```
Coordinate Reference System:
  User input: EPSG:4326
  wkt:
GEOGCRS["WGS 84",
    ENSEMBLE["World Geodetic System 1984 ensemble",
        MEMBER["World Geodetic System 1984 (Transit)"],
        MEMBER["World Geodetic System 1984 (G730)"],
        MEMBER["World Geodetic System 1984 (G873)"],
        MEMBER["World Geodetic System 1984 (G1150)"],
        MEMBER["World Geodetic System 1984 (G1674)"],
        MEMBER["World Geodetic System 1984 (G1762)"],
        MEMBER["World Geodetic System 1984 (G2139)"],
        MEMBER["World Geodetic System 1984 (G2296)"],
        ELLIPSOID["WGS 84",6378137,298.257223563,
            LENGTHUNIT["metre",1]],
        ENSEMBLEACCURACY[2.0]],
    PRIMEM["Greenwich",0,
        ANGLEUNIT["degree",0.0174532925199433]],
    CS[ellipsoidal,2],
        AXIS["geodetic latitude (Lat)",north,
            ORDER[1],
            ANGLEUNIT["degree",0.0174532925199433]],
        AXIS["geodetic longitude (Lon)",east,
```

```
            ORDER[2],
            ANGLEUNIT["degree",0.0174532925199433]],
    USAGE[
        SCOPE["Horizontal component of 3D system."],
        AREA["World."],
        BBOX[-90,-180,90,180]],
    ID["EPSG",4326]]
```

```r
st_crs(clinics_sf)
```

```
Coordinate Reference System:
  User input: EPSG:4326
  wkt:
GEOGCRS["WGS 84",
    ENSEMBLE["World Geodetic System 1984 ensemble",
        MEMBER["World Geodetic System 1984 (Transit)"],
        MEMBER["World Geodetic System 1984 (G730)"],
        MEMBER["World Geodetic System 1984 (G873)"],
        MEMBER["World Geodetic System 1984 (G1150)"],
        MEMBER["World Geodetic System 1984 (G1674)"],
        MEMBER["World Geodetic System 1984 (G1762)"],
        MEMBER["World Geodetic System 1984 (G2139)"],
        MEMBER["World Geodetic System 1984 (G2296)"],
        ELLIPSOID["WGS 84",6378137,298.257223563,
            LENGTHUNIT["metre",1]],
        ENSEMBLEACCURACY[2.0]],
    PRIMEM["Greenwich",0,
        ANGLEUNIT["degree",0.0174532925199433]],
    CS[ellipsoidal,2],
        AXIS["geodetic latitude (Lat)",north,
            ORDER[1],
            ANGLEUNIT["degree",0.0174532925199433]],
        AXIS["geodetic longitude (Lon)",east,
            ORDER[2],
            ANGLEUNIT["degree",0.0174532925199433]],
    USAGE[
        SCOPE["Horizontal component of 3D system."],
        AREA["World."],
        BBOX[-90,-180,90,180]],
    ID["EPSG",4326]]
```

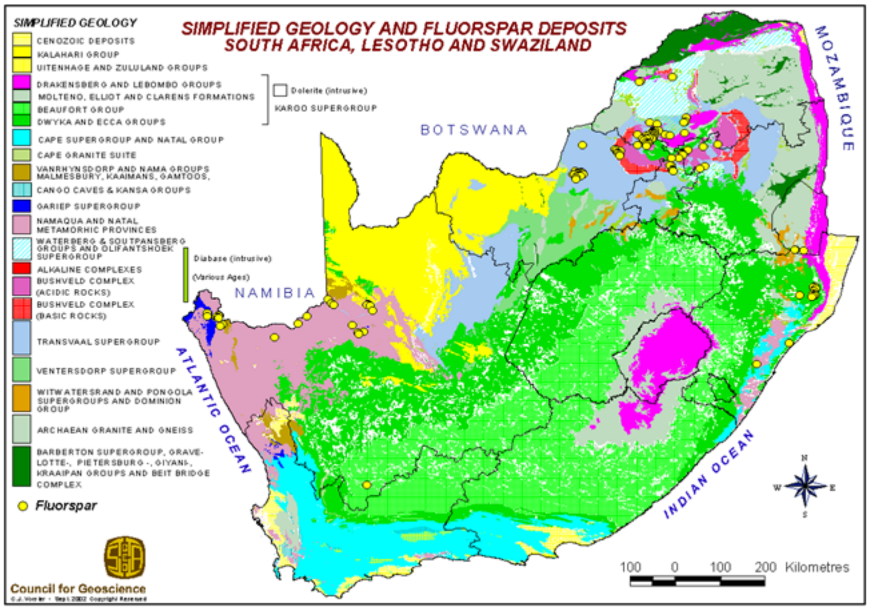
```r
class(clinics_sf)
```

```
[1] "sf"         "data.frame"
```

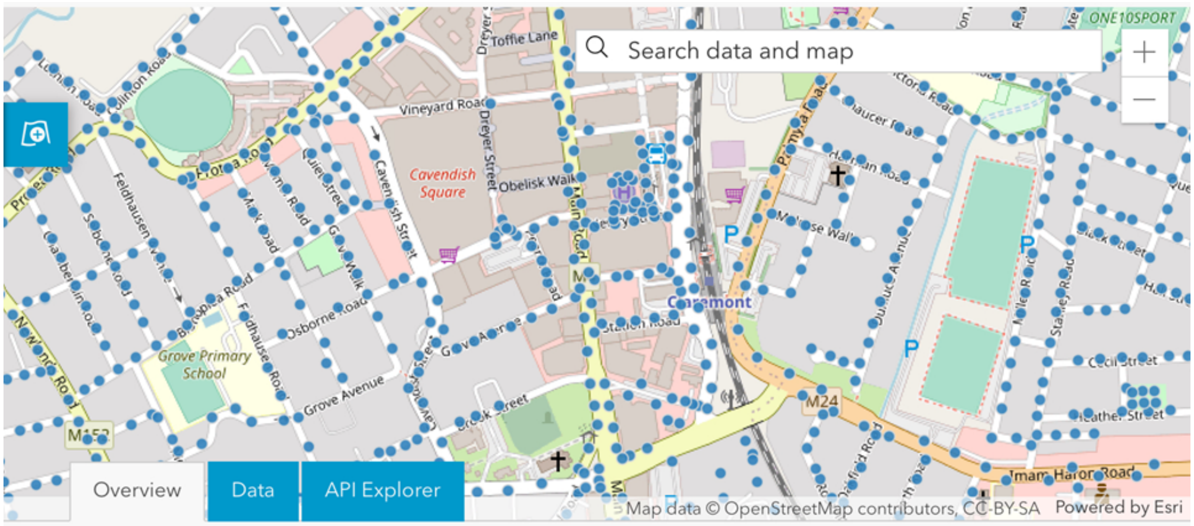
```r
summary(clinics_sf)
```

```
     LCTN               ATHY               NAME              CLASS          
 Length:149         Length:149         Length:149         Length:149        
 Class :character   Class :character   Class :character   Class :character  
 Mode  :character   Mode  :character   Mode  :character   Mode  :character  
     RGN                     geometry  
 Length:149         POINT        :149  
 Class :character   epsg:4326    :  0  
 Mode  :character   +proj=long...:  0  
```

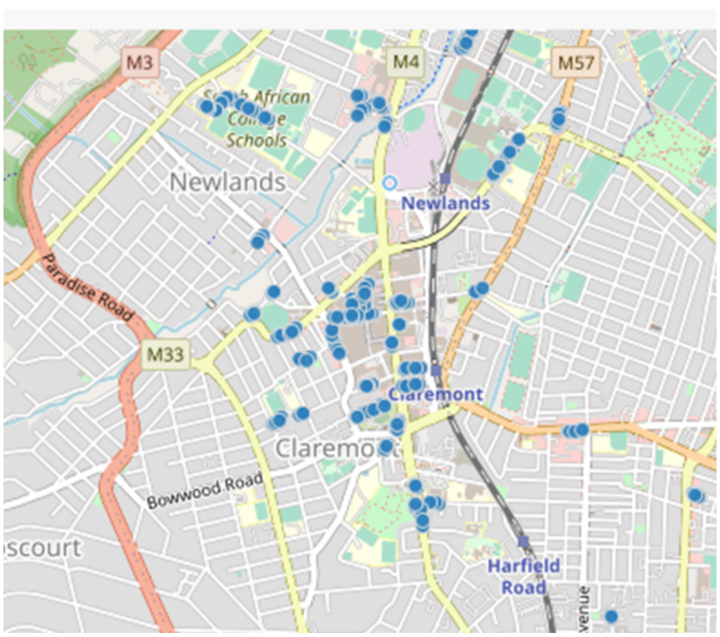
Example 5: Fluorspar



Example 6: Public Electricity Lighting



Example 7: On Street Parking



Data Resources for SPP for Cape Town Municipality

- https://odp-cctegis.opendata.arcgis.com/search?collection=Dataset
- https://odp.capetown.gov.za/datasets/wards/data
- Parks
- Wards shape file
- MyCiti Bus Stops
- Libraries

(Statistics by place)

Cape Town by documents: GeoDa Center

What is common in all these examples?

We have an event of interest (crime, trees, animals…etc.)

Point patterns are the locations of these events $\{\mathbf{s}_i\}$

Key difference from other spatial data: All points are known mapped pattern Selection bias

We know all the points in other words there are no missing values. We are only interested in where crime happens, not where it did not happen where it could have happened. (sounds like selection bias)

Our dataset may also include covariates—any data that we treat as explanatory, rather than as part of the 'response'. For the example of trees, the altitude is a covariate as well as soil type, acidity.